

Multimodal Retrieval-Augmented Generation for Chart and Document Understanding

Abstract

Large Language Models (LLMs) have demonstrated extraordinary capabilities in natural language understanding and generation; however, they remain fundamentally constrained by static parametric knowledge, leading to factual hallucinations and an inability to process dynamic, domain-specific information. Retrieval-Augmented Generation (RAG) emerged as the dominant paradigm to mitigate these limitations by dynamically grounding model outputs in external knowledge bases. However, traditional RAG architectures are strictly unimodal, treating knowledge as flat textual streams and severely degrading performance when applied to Visually Rich Documents (VRDs) such as scientific papers, financial reports, and technical manuals. These documents convey critical information through complex, interleaved modalities, including charts, tables, infographics, formulas, and intricate spatial layouts. Multimodal Retrieval-Augmented Generation (MRAG) represents a paradigm shift, extending retrieval and generation to span across text, images, and structural layouts. This survey presents a comprehensive, systematic analysis of the MRAG landscape specifically tailored for chart and document understanding. We propose a nuanced taxonomy of current methodologies, categorizing the field into foundational multimodal encoders, dense and sparse embedding architectures, late-interaction retrieval mechanisms, specialized chart-to-text translation frameworks, and advanced graph-based and agentic pipelines. Furthermore, this survey rigorously evaluates the current state of multimodal benchmarks, exposing the critical vulnerabilities of Multimodal Large Language Models (MLLMs), such as embedding collapse in visually homogeneous corpora and the catastrophic failure of numerical reasoning in unlabeled charts. By synthesizing current literature, architectural trade-offs, and empirical findings, this report outlines the most pressing challenges and charts a roadmap for future research in achieving holistic, scalable, and highly accurate multimodal document intelligence.

Introduction

The proliferation of information in the digital age is increasingly multimodal. In enterprise, scientific, and legal domains, critical knowledge is rarely confined to continuous prose; rather, it is embedded within Visually Rich Documents (VRDs) that integrate text, hierarchical layouts, tables, charts, and figures [1, 2]. While traditional Large Language Models (LLMs) and early Retrieval-Augmented Generation (RAG) systems have revolutionized text-centric knowledge acquisition, their reliance on Optical Character Recognition (OCR) to flatten documents into linear textual streams fundamentally destroys spatial, structural, and visual semantics [2, 3]. The OCR-based pipeline is inherently lossy: row-column bindings in tables are severed, the geometric relationships in organizational charts are lost, and the visual encodings of data plots

are entirely discarded during preprocessing [2, 4, 5].

To overcome these structural deficits, Multimodal Retrieval-Augmented Generation (MRAG) has emerged as a critical extension of the RAG framework [1, 3, 6]. MRAG integrates diverse data modalities—text, images, layout, and video—directly into the retrieval index and generation context, enabling Multimodal Large Language Models (MLLMs) to reason over holistic document representations [3, 7]. In an MRAG system, the retrieval module must align heterogeneous data types into a shared semantic space, distinguishing between fine-grained visual features, such as a specific trend line in a graph, and dense textual arguments [1, 8]. Subsequently, the generation module leverages MLLMs, instruction-tuned to synthesize these retrieved multimodal contexts, thereby producing responses grounded in both visual and textual evidence [3, 9].

Despite rapid advancements, MRAG for document and chart understanding introduces a unique set of challenges that diverge significantly from standard natural image retrieval [10]. Unlike natural images, which convey holistic, scene-level semantics, document pages are information-dense, highly structured, and often visually homogeneous [10, 11]. A financial invoice from one vendor may look geometrically identical to an invoice from another, causing traditional vision-language models pre-trained via contrastive learning (e.g., CLIP) to suffer from "embedding collapse"—a phenomenon where visually similar documents become indistinguishable in the dense vector space [11]. Furthermore, charts represent artificial symbolic mappings rather than natural visual features, requiring models to perform strict mathematical reasoning, coordinate mapping, and data derendering rather than mere object detection [12, 13, 14].

This survey comprehensively explores the intersection of MRAG and document/chart understanding. It maps the methodological evolution from foundational layout-aware pre-training to modern OCR-free paradigms. It dissects the current retrieval trilemma, balancing the efficiency of dense embeddings, the lexical precision of sparse embeddings, and the fine-grained cross-modal accuracy of late-interaction architectures [15, 16, 17]. Additionally, it evaluates the integration of structural and mathematical reasoning through multimodal knowledge graphs and layout-aware compilers [4, 5]. By systematically evaluating these methods, datasets, and inherent limitations, this paper aims to provide researchers with a rigorous foundation for advancing the next generation of multimodal document intelligence.

Methods

The architectural landscape of MRAG for document and chart understanding can be categorized into distinct methodological families based on how they process, index, and retrieve multimodal information. This section traces the flow of information, starting with foundational document encoders, progressing to the mechanics of visual document retrieval paradigms, detailing specialized techniques for charts and tables, and concluding with advanced structural enhancements such as graph-based and agentic RAG.

Foundational Encoders for Visually Rich Documents

The bedrock of any MRAG system is its ability to accurately encode the source documents. The evolution of document encoders has bifurcated into two primary philosophies:

OCR-dependent layout-aware models and OCR-free visual-language models.

Early successful attempts to process VRDs relied on combining off-the-shelf OCR engines with Transformer architectures [18]. LayoutLM and its successors, particularly LayoutLMv3, represent the pinnacle of this approach [18]. LayoutLMv3 utilizes a unified text-image multimodal Transformer that does not rely on complex Convolutional Neural Network (CNN) backbones, such as Faster R-CNN, for region feature extraction [18, 19]. Instead, it projects document images into discrete linear patches, analogous to Vision Transformers [18]. To capture spatial semantics, LayoutLMv3 introduces sophisticated 2D position embeddings. The bounding box of each OCR-extracted word, denoted mathematically by the absolute

coordinates (x_0, y_0, x_1, y_1) , is normalized and embedded alongside the traditional 1D sequence position [18, 19]. Unlike previous iterations that used word-level layout positions, LayoutLMv3 adopts segment-level layout positions where words belonging to the same semantic segment share identical 2D spatial coordinates, recognizing that words within a single block usually express the same semantic meaning [19].

The pre-training of LayoutLMv3 utilizes a tripartite objective: Masked Language Modeling (MLM), Masked Image Modeling (MIM), and a novel Word-Patch Alignment (WPA) objective [18, 19]. The WPA explicitly forces cross-modal alignment by predicting whether the corresponding image patches for a given unmasked text token have been corrupted, effectively teaching the model the fine-grained relationship between a word and its visual rendering [18].

While LayoutLMv3 is highly effective, it suffers from cascading errors originating in the OCR layer. If the OCR fails to detect a complex mathematical formula or a poorly scanned table cell, the downstream Transformer cannot recover the lost data. This vulnerability motivated the development of OCR-free, end-to-end image-to-text sequence models like Donut (Document Understanding Transformer) and Nougat (Neural Optical Understanding for Academic Documents) [20, 21]. Nougat utilizes a Swin Transformer encoder to extract features directly from the raw PDF page image and an autoregressive mBART decoder to output formatted Markdown or LaTeX [20, 21]. By skipping the OCR step, Nougat preserves the implicit visual structure of the document, excelling at translating highly complex inline mathematical equations and hierarchical reading orders that traditional OCR engines typically mangle [21, 22]. However, while models like Nougat are excellent for offline parsing, their autoregressive decoding is too computationally expensive for real-time retrieval indexing over millions of pages, necessitating alternative embedding strategies for large-scale MRAG [16].

Visual Document Retrieval (VDR) and Embedding Paradigms

In a multimodal RAG pipeline, the retriever is tasked with finding the most relevant document page across a massive corpus given a user query. This requires mapping visual documents into a latent space where they can be efficiently compared against the query.

A straightforward approach to VDR involves utilizing dual-encoder architectures powered by Vision-Language Models (VLMs) such as CLIP or SigLIP [11]. In this dense omni-modal paradigm, a document image is passed through a vision encoder, pooling the final hidden states into a single dense vector. The text query is similarly embedded, and retrieval is performed via Maximum Inner Product Search (MIPS) using approximate nearest neighbor

algorithms [15, 23]. Recent advances strive for "omni-modal" spaces, where text, image, audio, and document modalities are decoupled during specialist training but fused into a single unified embedding model [23]. Models such as Conan-embedding-v3 utilize a decouple-fuse-recover framework to prevent cross-modal optimization conflict, enabling a single embedding space without allowing dominant modalities to degrade the performance of specialized document retrieval pathways [23].

However, dense visual embeddings exhibit a critical vulnerability in document retrieval known as embedding collapse [11]. Natural images vary wildly in pixel distribution, making contrastive learning highly effective. Conversely, visual documents within a specific domain, such as thousands of corporate invoices or academic papers, are visually homogeneous. They share identical layouts, logos, and fonts, differing only in dense, localized text [11]. Standard vision encoders fail to differentiate these subtle semantic differences, collapsing the embeddings of distinct documents into a tightly packed cluster [10, 11]. To mitigate this, hybrid VL-RAG frameworks have been proposed that concatenate dense text embeddings with structural visual features, providing orthogonal semantic signals to disambiguate visually identical pages [11].

To capture the exact lexical matches that dense embeddings often miss, researchers have adapted learned sparse retrieval techniques for multimodal inputs. Models like DocRetriever and V-SPLADE generate layout-aware sparse embeddings without incurring OCR overhead [16, 17]. These models leverage the inherent language modeling (LM) head of a VLM. When a VLM processes a document image, its LM head produces a vocabulary-scale logit distribution reflecting the semantic importance of tokens detected in the image [16, 17]. By applying the SPLADE technique, which utilizes ReLU activation followed by log-saturation, the model mitigates the dominance of high-frequency tokens and extracts a sparse vector representing

the lexical content of the image [16]. Techniques such as word lemmatization and top- k logit selection compress this distribution into a lightweight representation [16]. V-SPLADE further relies on a gating mechanism where offline-generated captions guide the visual sparse vector, reinforcing dimensions supported by both modalities [17]. This allows the visual document to be indexed in a standard inverted index, enabling highly efficient exact search without neural query encoding at runtime [17].

Late Interaction Architectures

The trade-off between the coarse granularity of single-vector dense retrieval and the lexical rigidity of sparse retrieval has driven the adoption of Multi-Vector Late Interaction models. Originating with ColBERT in text retrieval, this paradigm computes similarities at the token level rather than pooling the entire document into a single vector [15, 24].

ColPali represents the state-of-the-art translation of late interaction into the visual document domain [15]. ColPali utilizes the PaliGemma-3B VLM, which combines a SigLIP vision encoder and a Gemma-2B language model [15]. Rather than pooling the output, ColPali applies a projection layer to map each image patch from the vision encoder into a 128-dimensional vector [15]. For a standard document image resized to high resolution, this yields a grid of

patches, generating a multi-vector representation $\mathbf{E}_d \in \mathbb{R}^{N_d \times 128}$ [15].

During retrieval, a text query is tokenized and embedded into $\mathbf{E}_q \in \mathbb{R}^{N_q \times 128}$. The similarity is calculated using the Late Interaction (MaxSim) operator [15]:

$$S(q, d) = \sum_{i=1}^{N_q} \max_{j=1}^{N_d} (\mathbf{E}_q^{(i)} \cdot \mathbf{E}_d^{(j)})$$

This mathematical formulation allows every individual token in the user's query to independently scan all patches of the document image to find its maximal semantic visual match [15]. Because the pooling step is omitted, ColPali preserves fine-grained layout, tabular, and graphical details, dramatically outperforming OCR-based pipelines on visually complex benchmarks [15, 25].

The primary drawback of ColPali is computational scalability. Storing over a thousand 128-dimensional vectors per page requires massive memory footprints, and computing MaxSim across millions of documents at runtime is prohibitively slow [24]. Recent research addresses this bottleneck through vector compression and pruning. Light-ColPali explores aggressive token merging strategies, successfully preserving 94.6% of the retrieval effectiveness while reducing the memory footprint to just 2.8% of the original model [26]. Furthermore, HPC-ColPali (Hierarchical Patch Compression) utilizes K-Means quantization to compress 128-dimensional patch embeddings into 1-byte centroid indices, achieving a 32x storage reduction. It also employs attention-guided dynamic pruning, using the VLM's attention weights to identify and retain only the top percentile of most salient visual patches, successfully discarding blank whitespace margins before storing the vectors in an index [27].

Chart and Table Understanding Methodologies

Within multimodal documents, charts and tables present the highest density of relational and numerical information. Chart understanding is a quintessential information fusion task; models cannot simply perceive a chart, they must interpret the artificial symbolic mapping of pixels to numerical values via coordinate geometries and legends [12, 13].

To integrate charts into the RAG generation phase, researchers have developed specialized derendering models. DePlot (One-shot visual language reasoning by plot-to-table translation) decomposes chart reasoning into a two-step pipeline [28]. First, the vision model translates the image of a plot into a linearized Markdown table. Second, this textual table is provided as the context to an off-the-shelf LLM [28]. By converting the visual geometry into a structural text grid, DePlot leverages the strong numerical reasoning capabilities of LLMs, achieving massive gains over direct visual-question answering on charts [28, 29].

MatCha (Math reasoning and Chart derendering pretraining) enhances this further. MatCha alters the pre-training objectives of the underlying Pix2Struct model by enforcing two complementary tasks: Chart derendering, forcing the model to decode the underlying rendering code or raw data table of a given chart, and Mathematical reasoning, rendering math problems as images and forcing the model to decode the solution [30]. Subsequent behavioral

analysis of MatCha revealed that MLLMs still struggle with structural visual context. Consequently, newer pre-training tasks in MatCha-v2 enforce Visual Structure Prediction, predicting chart types and legends, and Summary Statistics Prediction, forcing the model to predict the mean or maximum value directly from the pixels [31].

Tables require similarly precise structural modeling. Early models treated Table Detection (TD) and Table Structure Recognition (TSR) as separate object detection tasks. The Table Transformer (TATR), built on the DETection TRansformer (DETR) architecture, unified this by treating table rows, columns, and spanning cells as queryable objects [32, 33]. Recent extensions like POTATR (Page-Object Table Transformer) expand this to multi-page tables, utilizing relation heads to explicitly predict the semantic linkages between a row object on one page and the continued table header on the subsequent page [33]. Furthermore, models like UniTabDet focus on cross-lingual and cross-layout generalization, utilizing YOLO-based architectures to detect structurally diverse tables across varied global document standards [34].

Graph-Based Multimodal RAG

While vector databases excel at finding semantically similar document chunks, they fundamentally fail at multi-hop reasoning over structured, relational data spread across long documents. When text, tables, and formulas are flattened into independent vector chunks, the strict logical dependencies, such as the conditional applicability of a technical standard, are destroyed [4, 5]. Graph-based Multimodal RAG addresses this by compiling documents into Multimodal Knowledge Graphs (MKGs).

MegaRAG introduces a page-based, two-round approach for MKG construction [4]. Instead of independently extracting entities from segmented chunks, MegaRAG processes entire pages using MLLMs to extract text-to-text, text-to-figure, and figure-to-figure relations [4]. The initial graph acts as global guidance in a secondary refinement stage, where subgraphs are retrieved to provide context-aware inputs, progressively improving structural coherence and cross-modal grounding across full-length books [4].

In highly technical domains, accuracy depends strictly on operational logic embedded in tables and equations. SEM-RAG introduces a layout-aware compiler that bypasses flat text conversion [5]. It transforms tables into "condition graphs," where each cell is explicitly linked to its row header, column header, and predicates. It transforms formulas into "operator graphs," separating variables, constants, and functional dependencies tied to local symbol definitions [5]. To compress this massive graph for efficient indexing without relying on expensive LLM-based clustering, SEM-RAG utilizes Structural Entropy Minimization (SEM). This information-theoretic approach quantifies the expected coding cost of a random walk on the graph, algorithmically grouping dense local regularities into a compact hierarchy that supports factoid, relational, and global-summary retrieval [5].

Agentic and Pre-Retrieval Frameworks

The static retrieve-then-read paradigm is increasingly being replaced by dynamic, Agentic RAG frameworks. In these setups, LLM-driven agents evaluate their own uncertainty, execute iterative search queries, and filter multimodal evidence over multiple hops [2].

A persistent challenge in MRAG is the domain shift that occurs when a model trained on Wikipedia or natural images is deployed on proprietary, out-of-distribution corporate documents. PreMIR (Zero-shot Multimodal Document Retrieval via Cross-modal Question Generation) tackles this without requiring fine-tuning [35]. PreMIR uses an MLLM to actively generate synthetic cross-modal pre-questions (preQs) from the documents in the corpus prior to indexing [35]. By decomposing documents into visual, textual, and multimodal preQs, PreMIR maps complex documents into an expanded, token-level semantic space. At runtime, the user's query is matched against these preQs rather than the raw document embeddings, bridging the modality and domain gap entirely through generative augmentation [35, 36].

Methodology Family	Representative Models	Key Innovation	Retrieval Modality
Foundational Encoders	LayoutLMv3, Nougat	2D Layout embeddings; OCR-free autoregressive parsing	Text + Spatial Bounding Boxes
Dense Omni-Modal	Conan-embedding -v3	Decouple-fuse-rec over training for unified latent spaces	Text, Image, Video, Audio
Late-Interaction	ColPali, HPC-ColPali	Token-to-patch MaxSim; K-means compression pruning	Visual Patches
Lexical Sparse	V-SPLADE, DocRetriever	LM-head logit extraction (ReLU + Log-saturation)	Visual-to-Lexical Sparse
Chart/Table Derendering	DePlot, MatCha, POTATR	Plot-to-table translation; Math reasoning pre-training	Image-to-Text / BBox Detection
Graph-Based RAG	MegaRAG, SEM-RAG	Operator graphs; Structural entropy minimization	Multimodal Knowledge Graph

Agentic / Pre-Retrieval	PreMIR	Cross-modal pre-question generation for domain shift	Synthetic Textual PreQs
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Datasets and Benchmarks

The transition from unimodal to multimodal RAG has necessitated the creation of complex, multi-page, reasoning-intensive benchmarks. Early Visual Document Understanding (VDU) datasets like DocVQA or TextVQA were restricted to single-page, extractive answers, failing to evaluate the multi-hop retrieval and contextual synthesis required in real-world MRAG deployments [37, 38]. Recent datasets are categorized by their structural focus and corpus scale.

Chart and Structural Benchmarks

Evaluating models on structured data requires datasets that decouple textual semantic understanding from pure visual-mathematical reasoning [12]. The Chart-MRAG Bench, developed via the CHARGE framework, features 4,738 expert-validated QA pairs across eight domains. It specifically challenges models on information density, establishing that 70% of charts in the dataset contain more than eight critical keypoints, severely testing an MLLM's ability to track dense numerical data [39].

GRAFT (GRaPH and Table Reasoning for Textual Alignment) addresses the lack of structural controllability in natural datasets by using Python visualization libraries to programmatically generate charts and tables [40]. This ensures strict control over data semantics and requires models to output structured formats, such as JSON or YAML, based on complex analytical tasks like proportion estimation, trend identification, and anomaly detection [40]. Similarly, while traditional DocVQA requires text answers, BBox-DocVQA forces the MLLM to provide the spatial bounding box of the evidence. This unique annotation isolates the model's visual grounding capability from its parametric memorization, preventing models from guessing correct answers without actually reading the document [41].

Multi-Page and Domain-Specific Document Benchmarks

To simulate realistic RAG environments, benchmarks must include thousands of pages to evaluate retrieval precision and generation grounded in multiple pages [2, 10]. ViDoRe V3 evaluates full end-to-end multimodal RAG pipelines over visually rich corpora [37]. It includes 26,000 pages and 3,099 human-verified queries across six languages. The queries span a dual-axis taxonomy, contrasting open-ended, extractive, and numerical queries against question, keyword, and instruction formats [37]. The benchmark proves that late-interaction visual retrievers consistently outperform OCR-text retrievers [37, 38].

UniDoc-Bench features 1,600 synthesized multimodal QA pairs grounded across 70,000 highly relevant PDF pages [42]. UniDoc-Bench leverages an entity-overlapping knowledge graph to generate multi-hop questions that absolutely require the fusion of text, images, and tables to

answer correctly, exposing the severe inadequacy of relying on a single modality [42]. MMESGBench provides a domain-specific dataset for Environmental, Social, and Governance (ESG) documents, notorious for heterogeneous layouts and dense financial charts [43]. The dataset highlights that while MLLMs outperform text-only models by 60% on layout/chart queries, smaller text-only models dangerously inflate accuracy on unanswerable negative samples due to hallucination [43]. Finally, Invoice Haystack is specifically designed to stress-test embedding collapse in dense retrieval models [11]. It features 1,500 visually identical, anonymized invoices, forcing the retriever to rely on fine-grained textual variances rather than global visual similarity [11].

Benchmark Name	Target Modality	Key Feature / Innovation	Scale / Document Type
Chart-MRAG Bench	Charts, Text	High information density keypoint extraction	4,738 QA pairs, 8 domains
GRAFT	Synthetic Charts, Tables	Programmatically generated, JSON/YAML outputs	Analytical tasks (trend, proportion)
BBox-DocVQA	Document Layout	Enforces spatial bounding box grounding	Bounding-box-grounded QA
ViDoRe V3	End-to-End MRAG	Dual-axis query taxonomy, cross-lingual	26,000 pages, 3,099 queries
UniDoc-Bench	Text, Image, Table	Entity-overlapping knowledge graph QA	70,000 PDF pages, multi-hop
MMESGBench	Domain-Specific (ESG)	Tests hallucination on unanswerable queries	Financial and sustainability PDFs
Invoice Haystack	Homogeneous Documents	Stress-tests "Embedding Collapse"	1,500 identical anonymized invoices

Challenges and Prospects

The rapid evolution of MRAG has unlocked comprehensive document intelligence, but it has simultaneously exposed systemic vulnerabilities in MLLM architectures, retrieval scaling, and cross-modal alignment [1, 2, 8].

The OCR-Free vs. OCR-Based Paradox

The field is currently caught in a paradox regarding data extraction [2, 10]. OCR-based pipelines are computationally cheap and highly scalable for massive RAG indexes, but they irreparably destroy document layout, table structures, and visual context [1, 3]. Conversely, OCR-free VLM approaches preserve holistic spatial semantics but introduce massive computational overhead [10]. For example, rendering every page as a multi-vector sequence increases index storage requirements by orders of magnitude compared to plain text [15, 26]. Future systems must resolve this paradox, likely through adaptive architectures where lightweight OCR handles standard text, while expensive visual embeddings are dynamically triggered only for structurally complex regions [2, 44].

Embedding Collapse and Semantic Homogeneity

As demonstrated by the Invoice Haystack benchmark, dense visual embeddings pre-trained on natural images via contrastive learning fail catastrophically on homogenous corporate documents [11]. Because these models are tuned to distinguish drastically different objects, they struggle to distinguish an invoice from one vendor from an identically formatted invoice from another when the only difference is the text string of the vendor name [10, 11]. Addressing embedding collapse requires novel contrastive training objectives that heavily weight local lexical discrepancies over global structural similarity, or the universal adoption of hybrid dense-sparse retrieval mechanisms [11, 16].

The Efficiency and Scalability Constraint

Late interaction models achieve state-of-the-art retrieval accuracy by computing MaxSim between every query token and every document image patch [15]. While highly accurate, the computational complexity at runtime is prohibitive for enterprise-scale corpora containing millions of documents [24, 26]. The prospect of MRAG scalability relies on algorithmic compression. Techniques such as Structural Entropy Minimization, K-means quantization of patch vectors, and VLM-attention-guided pruning offer promising pathways [5, 27]. Additionally, the repurposing of VLM logit distributions into lexical sparse arrays presents a highly viable solution for leveraging existing inverted index infrastructure without sacrificing visual grounding [17].

Cross-Modal Alignment and Granular Reasoning

A critical deficiency remains in how MLLMs reason over specific data visualizations. When data labels are removed from bar charts or scatter plots, the accuracy of leading MLLMs plummets, indicating an inability to perform true geometric coordinate mapping [13, 14]. Current models rely too heavily on implicit OCR text cues embedded within the image rather than true

multimodal spatial reasoning [13, 14]. The future of chart understanding in MRAG relies on two-stage cognitive pre-training: teaching the model the foundational geometry of coordinate spaces before training it to map those visual encodings to semantic data [13]. Furthermore, integrating explicit mathematical operator graphs directly into LLM prompting will be necessary to ensure absolute factual reliability in mission-critical applications [5].

Conclusion

Multimodal Retrieval-Augmented Generation represents a vital evolutionary leap in artificial intelligence, transitioning systems from blind textual processors to holistic, visually grounded reasoning engines capable of navigating the complex multimodal reality of human knowledge. By mitigating the destructive constraints of traditional OCR pipelines, MRAG unlocks the semantic wealth hidden within charts, hierarchical tables, infographics, and spatial layouts. This survey has delineated the core methodologies driving this shift, from foundational layout-aware pre-training and specialized chart-to-table translation to the nuanced paradigms of visual document retrieval via dense, sparse, and late-interaction architectures. While models demonstrate the immense potential of patch-level interaction and structural graph compilation, the field faces significant hurdles. Embedding collapse in homogeneous documents, the computational bottleneck of multi-vector scaling, and the persistent cognitive failure of MLLMs to execute strict mathematical reasoning on unlabeled charts remain critical frontiers. Overcoming these challenges will require a convergence of adaptive retrieval granularity, algorithmic compression, and multi-stage geometric pre-training. As these systems mature, MRAG will fundamentally redefine document intelligence, enabling precise, verifiable, and comprehensively grounded automated reasoning across every domain of digital information.

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